



Integration of antifouling and foul-release moieties for optimizing the performance of PEG-silicone coatings

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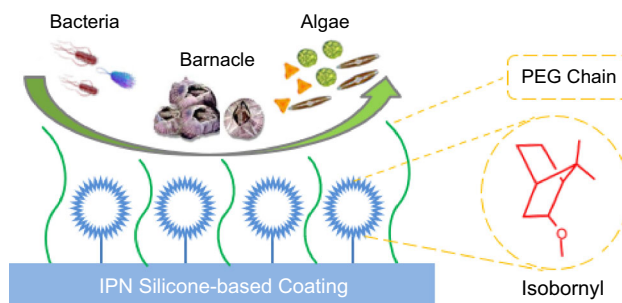
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Abstract Antifouling coatings are critical in preventing the surface of sea vessels being heavily fouled by bio-organisms. In this work, we attached biocidal moieties (Isobornyl methacrylate, IBOMA) to the side chains of the polyacrylic polymer to form the antifouling polymer (AFP). Then, the hydrophilic polysiloxane (HP) was prepared through the polyaddition of polyethylene glycol (PEG) and alkoxy silane onto the main chain of the hydrogen silicone oil. Finally, the hydrophilic fouling release coatings (FRCs) were formed by the reaction or winding of antifouling polymer and HP polymer with an interpenetrating polymer network (IPN) method (IPN-FRCs). The effects of the content of AFP and HP on the adhesion, surface roughness, and antifouling performance of the applied coatings on substrate are thoroughly studied and revealed in details through different analytic techniques. The IPN-FRCs exhibit high adhesion strength to the substrate, and the surface roughness is greatly reduced when submerged in seawater. Furthermore, the IPN-FRCs have high performance against seawater bio-organisms, such as barnacles, during both in-lab and marine field tests when right polymer ratios

were chosen. Thus, the novel FRCs may have great potential to be applied for marine vessels in line with their low-cost in raw materials and high performance of the final products.

Graphic abstract

The silicone-based antifouling polymer coatings show high resistance to the foulants of marine bio-organisms.



Keywords Antifouling coatings, Fouling release coatings, Silicone coating, IPN method

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Introduction

The settlement and development of marine organisms on the surface that is immersed in seawater is known as fouling or biofouling.¹ Such undesirable (often inevitable) colonization by marine organisms may cause detrimental impacts for the vessels with increased friction forces due to higher roughness, accelerated corrosion, and increased fuel consumptions, which leads to significant financial loss for the naval industry.² Many antifouling coatings have been developed to tackle this worldwide problem, and the common ones include heavy metals or toxic coatings such as tributyl

tin (TBT)-based compounds which could have adverse effects on many aquatic lives due to their high toxicity.^{3,4} Consequently, there are many restrictive legislations in place worldwide for the use of toxic antifouling coatings, promoting the fast development of nontoxic, environmentally friendly fouling release coatings (FRCs).⁵

FRCs are biocide-free and mainly silicone-based coatings which can reduce the adhesion between marine organisms and the vessel surface, and as a result, the fouling can be largely removed during navigation.^{6–9} Currently, the commercially available FRCs appear to be effective for high-speed vessels, and the reported lifetime for these FRCs is typically more than 5 years. However, FRCs are not so effective for the vessels in low-speed navigation. This is mainly because the ability of the silicone elastomers to resist fouling originates from their low surface energy or low elastic modulus, and the marine microorganisms can weakly adhere to the surface.^{10,11} Thus, external shearing force is required to detach fouling organisms from the silicone surface in static immersion or low speed navigation.^{12,13} On the other hand, it appears that some slime that contains bacteria and diatoms can accumulate on the silicone surface, which is hard to be removed during navigation.^{7,14} Therefore, the improvement for the fouling release ability of FRCs is necessary, particularly that on static or low speed conditions.¹⁵

One possible approach to improve the antifouling ability of silicone is to introduce addition of antifoulants.¹⁶ However, most antifoulants cannot steadily release into the seawater due to the silicone not being soluble. On the other hand, the released antifoulants may have a detrimental effect on the aquatic life, particularly those nontargeted marine organisms due to the toxicity of the antifoulants.¹⁷ Although chemically immobilized biocidal moieties such as organic antifoulants,^{18,19} quaternary ammonium salts,²⁰ zwitterions,^{21–23} and amphiphile²⁴ on silicone elastomers are able to avoid their release into the seawater, most of such biocidal moieties are firmly trapped in the bulk, leading to a poor fouling resistance. To possibly solve the problem, the biocidal moieties could be anchored on the coating surface, which results in reasonable antifouling performance for the silicone elastomer.²⁵ However, the silicone elastomers tend to cover the biocidal moieties during the formation of coating due to its low surface energy, making it difficult for this technique. Furthermore, due to either the immobilization or grafting technique, the processing of making such coatings is very complicated, limiting their application at large scales.

In this work, the biocidal moieties (Isobornyl methacrylate, IBOMA) were introduced to the side chains of the polyacrylic polymer resin. Then, a hydrophilic polysiloxane was prepared through the polyaddition of polyethylene glycol (PEG) and alkoxy silane onto the main chain of the hydrogen silicone oil. Finally, the hydrophilic fouling release coatings (FRCs) were formed by the reaction or

winding of antifouling polymer and HP polymer with an interpenetrating polymer network (IPN) method (IPN-FRCs). The effects of the content of acrylic resin and hydrophilic polysiloxane on the adhesion, surface roughness, and antifouling performance of the coatings are thoroughly investigated and revealed in details. It is apparent that the IPN-FRCs exhibit adequate adhesion strength to the substrate, and the surface roughness is greatly reduced when submerged in seawater. Furthermore, these IPN-FRCs have high performance against seawater bio-organisms, such as barnacles, during marine field test when right polymer ratios were chosen. Therefore, the novel FRCs have considerable applicability in line with their low-cost in raw materials and high performance in the final products.

Experimental

Materials

Isobornyl methacrylate (IBOMA), n-butyl methacrylate (BMA), Azo-iso-butyronitrile (AIBN), 2,2'-Azo-bis-(2-methylbutyronitrile) (AMBN), Benzoyl peroxide (BPO), 1-Dodecanethiol, Tetraethoxysilane (TEOS), di-n-butyltindilaurate (DBTDL), Vinyltriethoxysilane (VTEO), Xylene (Xy) were received from Xiya Regent Chemical Co. (Shandong, China). Hydrogen silicone oil (Si–OH) were received from Shandong Dayi Chemical Co. (Shandong, China). Karstedt's catalyst (0.1 M), Polyethylene glycol allyl ether (PEGAE, Mn = 500) were received from Shanghai Jinjinle Industrial Co. (Shanghai, China). PEGAE was stored over dry molecular sieves. Xy was refluxed and then distilled over sodium. Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2013).

Preparation of antifouling coat

Preparation of antifouling polymer (AFP) The AFP was synthesized by polyaddition as illustrated in Scheme 1. In a temperature-controlled reaction vessel equipped with a condenser, a thermometer, a dropping funnel, a mechanical stirrer, and argon inlet, Xy (50 g) and AMBN (0.5 g) are charged and heated to 100°C under stirring. From the dropping funnel, a transparent mixture consisting of IBOMA (150 g, BMA (150 g), Xy (236.0 g), Dodecanethiol (3.0 g), AIBN (1.5 g) and AMBN (1.5 g) is dropped at a constant rate over a period of 6 h. After the completion of dropping, Xy (10 g) and BPO (0.5 g) are further dropped over a period of 30 min and then heated to 110°C with stirring for 3 h. The molecular weights for the terpolymer resins were determined by gel permeation chromatography (GPC, Malvern, ViscOTEK TDA) measurements in THF. The calibration was based on polycal PMMA standards (Mn = 51092, PDI = 1.038; Mn = 95934, PDI = 1.838) were used to the calibration curve.

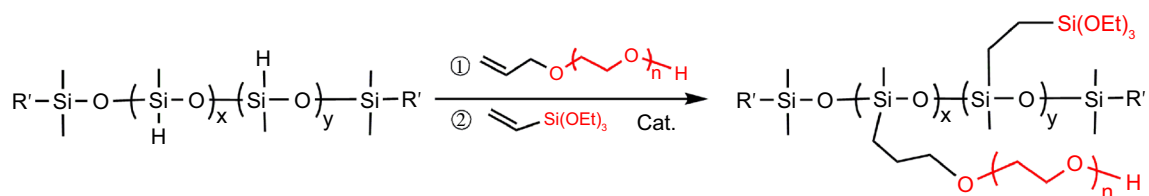
Table 1: Compositions of IPN-FRCs.

Preparation of alkoxysilane modified hydrophilic polysiloxane (HP) Alkoxysilane modified HP was synthesized by polyaddition as illustrated in Scheme 2. In a temperature-controlled reaction vessel equipped with a condenser, a thermometer, a dropping funnel, a mechanical stirrer and argon inlet, Karstedt's catalyst (27 μL), Si-OH (25 g) and Xy (50 g) are charged and heated to 75°C under stirring. From the dropping funnel, a transparent mixture consisting of PEGAE (55 g) and Karstedt's catalyst (15 μL) are dropped at a constant rate over a period of 45 min. After the completion of dropping, and then keeping this temperature with stirring for 3 h, VTEO (11 g) and Karstedt's catalyst (15 μL) are dropped at a constant rate over a period of 15 min, and then keeping this temperature with stirring for 4 h, the desire HP was obtained.

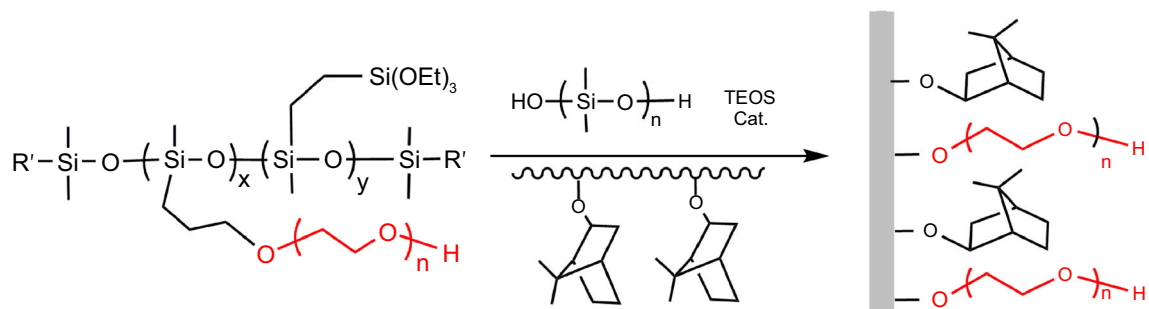
IPN-FRCs characterization

Brucker, USA) in the transmission mode in the range 4000–500 cm^{-1} at room temperature with a resolution of 2 cm^{-1} and accumulation of 16 scans. The swelling of each coating was determined by a gravimetric method. Each coat was casted on a glass slide and its initial dry weight (W_0) was recorded. The coated glass side was submerged in artificial seawater which was changed every second day. The glass slide was removed from the seawater on a designated day and rinsed with water. The excess water mainly on the surface was removed and the weight of the coat was recorded (W_t , wet). To estimate the dry weight (W_t , dry), the glass slide with coat was dried at 25°C for 1 day and then at 100°C for 6 h. This procedure was repeated on day 7, 14, 22, 28, 35, 42, 50, 62, 72, and 90 to evaluate the swelling of coats as a function of time. The weight difference between W_t , wet and W_0 reflected the swelling of each coating. Each coating was measured at least three times to get the average value. Contact angle measurements were performed on a contact angle measuring instrument (SINDIN SDC-200S, Dongguan Shengding Precision Instrument Co. LTD., Dongguan, China) with an automatic dispenser. The contact angle of each sample was measured after it was taken out from the seawater, tilted and placed for 30 min, and dried naturally on the surface. A droplet of Milli-Q water was placed on the surface of each sample; then the liquid was withdrawn immediately. The average of right and left contact angles was measured using Young-Laplace fitting method. Contact angles at four different spots on the surface were taken to get the average value. Surface roughness was conducted on an ultra-depth-of-field electron microscopy (KEYENCE VHX-6000, Keyence Corporation, Osaka, Japan). Adhesion measurements were carried by the adhesion tester (Elcometer-106, UK). The algae attachment experiments were carried out and fluorescence photographs were taken by an inverted fluorescence microscope (Leica DMI8, Leica Mikrosysteme Vertrieb GmbH, Wetzlar, Germany).

The antifouling test is conducted in accordance with the national experimental standard for marine antifouling coatings at shallow sea. On a $300 \times 250 \times 3 \text{ mm}^3$ steel substrate, epoxy zinc rich primer coat was firstly applied and then followed by the as-prepared antifouling coating. Three sets of parallel samples were used for each coating to test



Scheme 2: Synthetic routes of alkoxy silane modified HP.



Scheme 3: Synthetic route for IPN-FRCs.

their antifouling capability at different depths in the ocean. Four corners of the steel plate with coating were bolted on the hanger which was placed on the experimental floating raft in the testing sea area, located in the offshore of Louman's sheltered bay, Zhoushan City, China (30°01' N, 122°06' E) from June 13, 2019. The velocity of seawater was lower than 5 m/s, the test depth was 0.3–2.0 m, and the main fouling organism attached to the board was barnacle.

Results and discussion

FTIR analysis

Figure 1 shows the FTIR spectra of two representative FRCs at different timeframes (2 h and 24 h). The characteristic stretching vibration around 1650 cm^{-1} for the C=C double bond is not evident, indicating the completion of copolymerization reaction in the acrylic resins. The characteristic FTIR peaks for these samples are assigned as the following: the peaks at 2963, 2907, and 2873 cm^{-1} can be attributed to the stretching vibration peaks of the saturated C–H bond; the 1415 cm^{-1} peak is the in-plane bending vibration peak of C–H bond or –OH bond; the sharp and strong peak at 1260 cm^{-1} is the symmetrical deformation vibration for Si–CH₃; the broad peaks at 1100, 1020, and 796 cm^{-1} are the stretching vibration for Si–O bond, and Si–O–Si bond and Si–C bond, respectively. These IR observations are also in good agreement with our previous works.²⁶

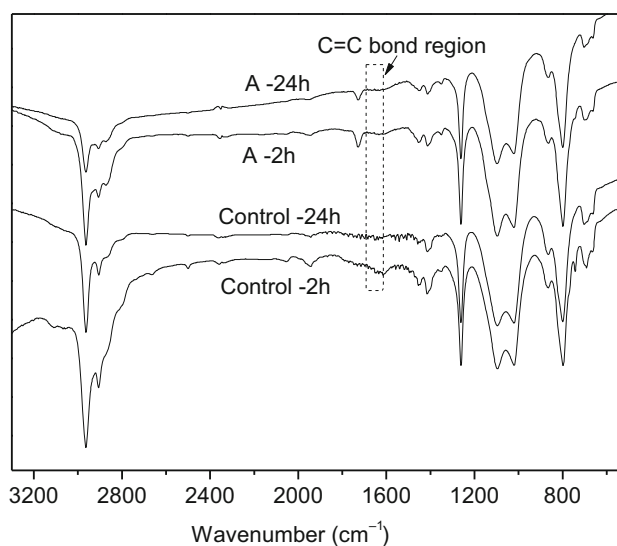


Fig. 1: FTIR spectra of two representative FRCs at different curing periods.

Swelling rate analysis

The rate of swelling of these IPN-FRCs immersed in water is shown in Fig. 2, which suggests that the samples swelled at a steady rate initially and then reached plateau in about 50 days. This may imply that the migration of hydrophilic polysiloxane groups onto the surface took about 50 days to reach equilibrium. The Control, A, D, and E samples had similar swelling rates, which could be due to the fact that they had the same ratio of hydrophilic polysiloxane. On the other hand, the weight of acrylic resin had little effect

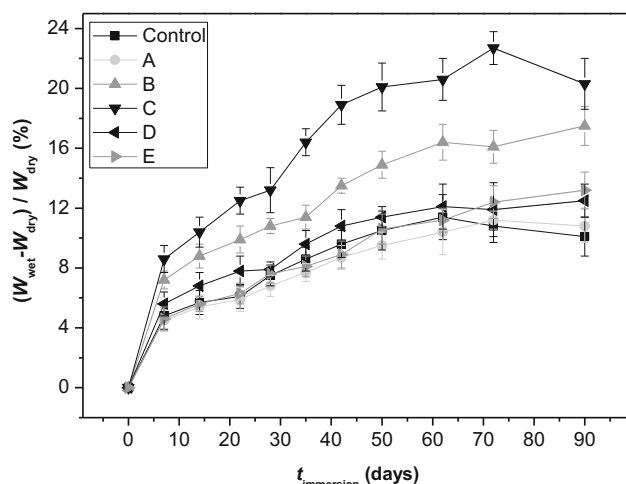


Fig. 2: Swelling rate of IPN-FRCs as function of time.

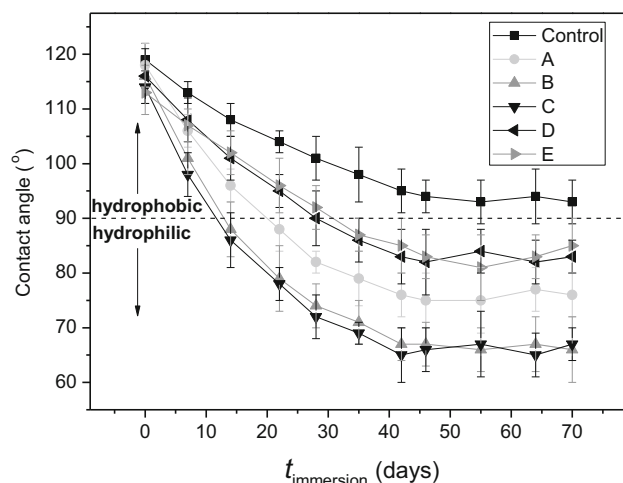


Fig. 3: Water contact angles of IPN-FRCs submerged in seawater for different times.

on the swelling of the coat. In the samples of A, B, and C, the swelling rate increases readily with increasing the hydrophilic polysiloxane, implying that the increase in the amount of polysiloxane could effectively increase the hydrophilicity of IPN-FRCs.

Contact angle measurement

The degree of water contact angle (θ) is used to probe the hydrophobicity of the solid FRCs: if $\theta > 90^\circ$, it is a hydrophobic surface; if $\theta < 90^\circ$, it is a hydrophilic surface. The FRCs were immersed in artificial seawater for different time periods and water contact angles were measured for these immersed samples as shown in Fig. 3. The contact angle is decreasing for all samples after immersion which is similar to other amphiphilic polymer coatings reported previously.²⁷ The increased hydrophilicity of the samples which were submerged in seawater is mainly due to the PEG chains being flexible

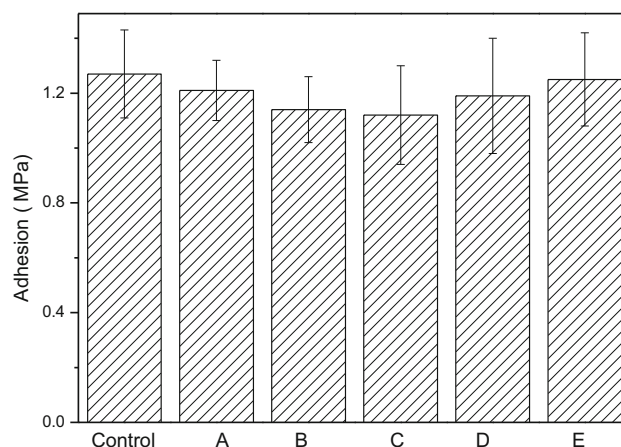


Fig. 4: Adhesion force of the coatings on the steel substrate.

and linear, and the hydrophilic C–O–C groups migrating onto the surface. It seems that the decrease in the water contact angle reaches a plateau in about 45 days, meaning the re-arrangement of the polymer chain is completed. This is in good agreement with the swelling results in the previous section. It is also noticeable that the presence of AFP causes the decrease in contact angle, possibly due to the large Isobornyl groups being able to leave easily, which leads to the increase in the hydrophilicity of the antifouling coatings.

Adhesion force of the coatings

The adhesion between the FRCs and the substrate is an important factor affecting the applicability of the coat. Figure 4 shows the adhesion (pull-off strength) of all six samples on the metal substrate, illustrating that all samples had adhesion force greater than 1.1 MPa, which is apparently above the commonly accepted 1.0 MPa benchmark required for the application of IPN-FRCs in the marine environment. From this figure, it appears that the adhesion slightly increases from Sample B to E when the antifouling polymer amount increases. This is because the hydrogen bonding density is increased in these samples as more acrylate content is used, leading to an increase in adhesion. However, on the other hand, the PEG content seems to have little effect on the adhesion.

Surface roughness

The anti-fouling properties the IPN-FRCs are significantly affected by their (both dry and wet) surface roughness which was measured on an ultra-depth-of-field electron microscopy as shown in Fig. 5 (and Fig. S1, Supporting Information). The surface roughness of the dry surface for all samples is around 13 μm , showing little difference among them. However, the roughness of these coatings is greatly reduced to

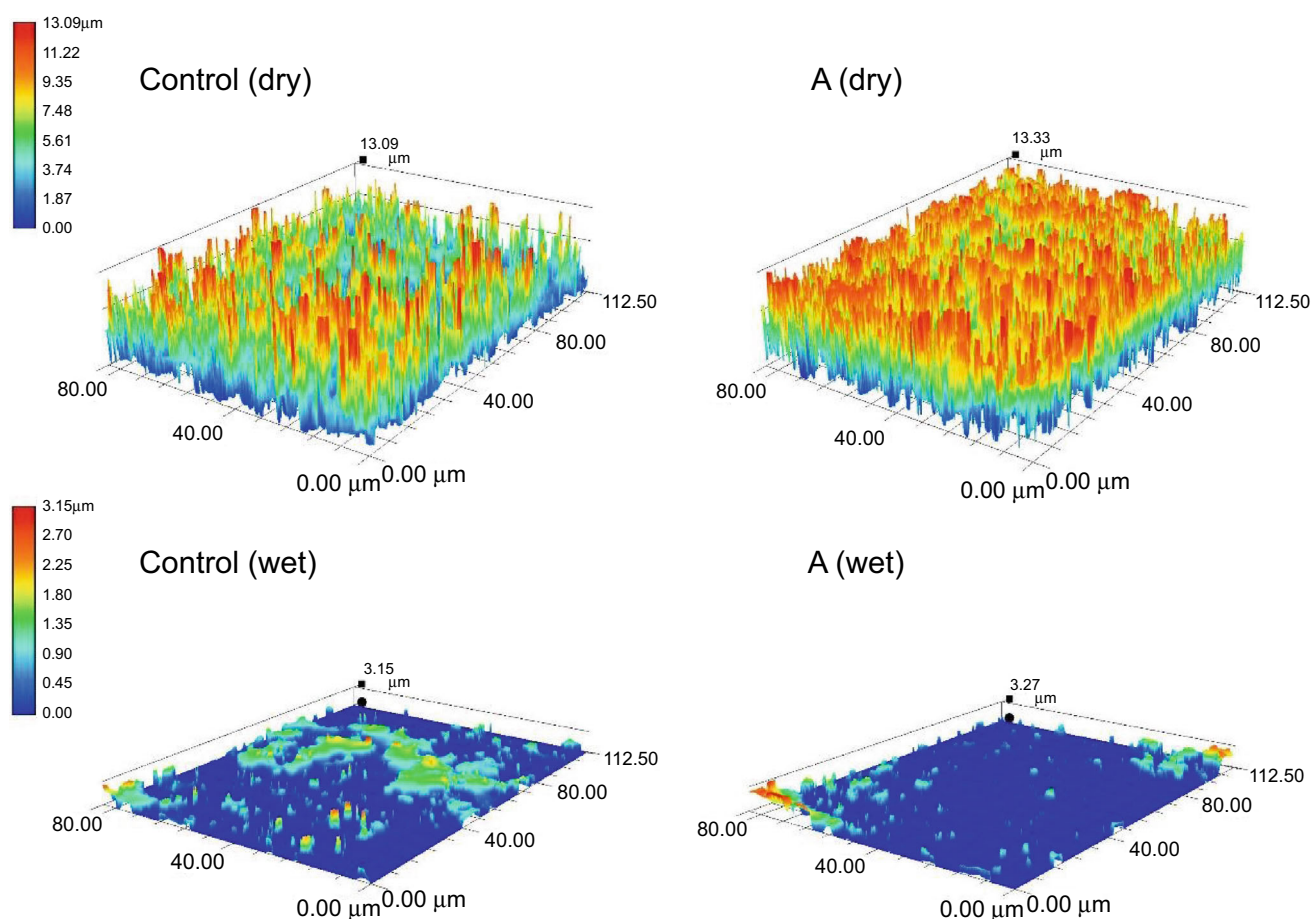


Fig. 5: Surface roughness images of Control and A resins of dry and wet surfaces taken from an ultra-depth-of-field electron microscopy.

3–5 μm in seawater. For instance, the surface roughness for the Control and Sample A, which has relatively small amount of AFP, are 3.15 μm and 3.27 μm , respectively, meaning smaller amount of AFP did not affect the roughness. Compared to Sample A, the roughness of the wet surface increases from B (3.68 μm) to C (5.60 μm) with increasing the hydrophilic polysiloxane, suggesting that further increasing the hydrophilic polysiloxane content can result in an increase in the roughness. This is mainly because the increased hydrophilic portion can lead to a higher swelling rate as illustrated in the previous swelling experiments. On the other hand, when the antifouling acrylic resin content is increased, the surface roughness is also increased from 4.16 μm for Sample D to 5.18 μm for Sample E. The possible explanation for this phenomenon is that the increased rigid parts in the acrylic resin may restrain the softer and smoother hydrophilic polysiloxane chains.²⁸ Nonetheless, the wet PEG-silicone IPN polymer coatings (submerged samples) have greatly smoother surfaces, reducing the possibility of initial attachment of fouling organisms.²⁹

Fluorescence microscopy analysis for algae inhibition performance

All samples were exposed to microalgae suspension for 48 h and their algae inhibition performance was qualitatively analyzed by examining the sample surface under a fluorescence microscope as shown in Fig. 6. In the control sample without acrylic resin, the settlement fraction of algae on the surface is 0.95% which reduces to 0.25% for Sample A, 0.22% for Sample B, and 0.16% for Sample C with increasing the hydrophilic polysiloxane content. However, when the AFP is largely increased (Samples D and E), the algae settlement fraction is in fact increased (meaning a poorer ability against algae settlement). This trend is closely similar to that of the surface roughness discussed in the previous section. This result has suggested that there is an interplay between the ratio of AFP resin and hydrophilic polysiloxane resin in the final applied coating and appropriate ratios of these resins need to be chosen in order to achieve high performance coatings. For the sake of comparison, we also applied commercial epoxy resin onto a blank

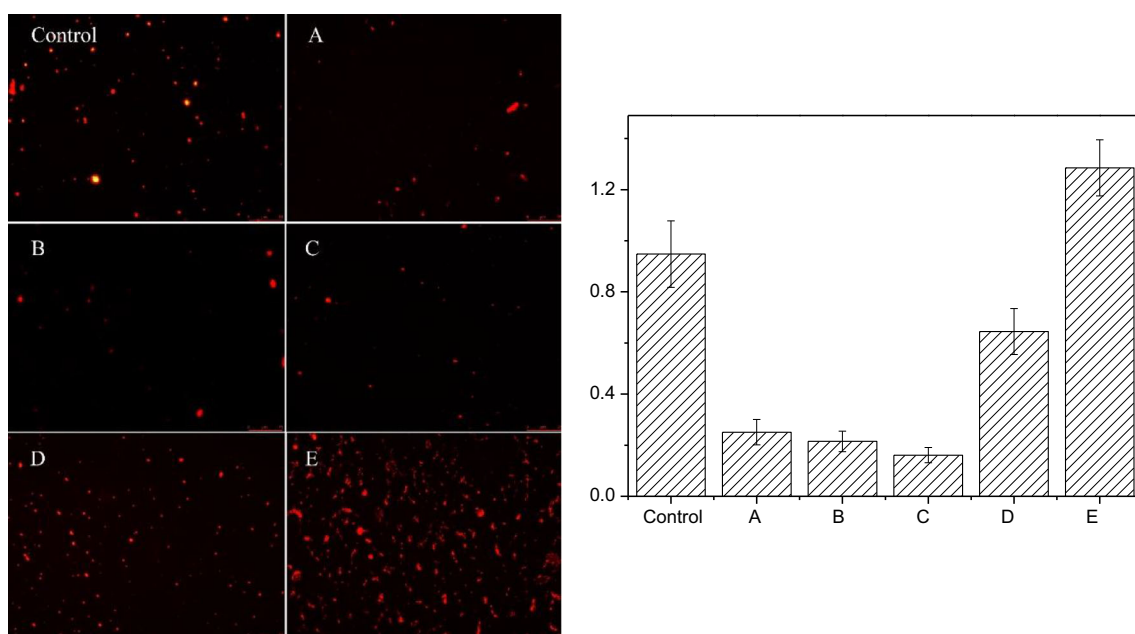


Fig. 6: Algae inhibition experiments. (left) Fluorescence images of the IPN-FRCs after inhibition test; (right) Inhibition rate of algae calculated via the attachment areas.

panel and it was tested for algae settlement which was 3.9%, much higher than the ones with prepared FRCs. This illustrates that our samples have excellent antifouling performance from the viewpoint of against algae settlement.

Marine field test

The antifouling performance evaluated in the laboratory conditions may not be adequate to reflect their capability in real marine environment. Thus, we examined the panels with different coatings in marine field in which these samples were submerged for 70 days in order to assess their applicability. Figure 7a shows the images of surfaces from the samples after 70-day marine field test. The panel coated by commercial epoxy resin had a large amount of fouling bio-organisms, barnacles in particular. The application of AFP onto the panels reduces the fouling features. We counted the number of barnacles on each panel as shown in Fig. 7b. It is not surprising that all samples had been fouled after 70 days in seawater, but it is confirmed that the samples coated with AFP had fewer number of barnacles attached on the surface, indicating that the AFP coating was indeed effectively protecting the steel panels. On the control sample (without acrylic resin), the number of barnacles is found to be 52 and this number reduces to 36 for Sample A containing AFP resin, illustrating that the AFP plays a role against bio-organism settlement, as expected. However, when there is an increase in the content of hydrophilic polysiloxane (B and C) or AFP, the antifouling ability appears to be more or less reduced. This is, on one

hand, suggesting that appropriate ratios of the components are critical to achieve high performance, and on the other hand, it has shown that the complexity in real marine applications.

Antifouling mechanism

Figure 8 shows the antifouling mechanism of the IPN-FRCs applied on the substrate. The isobornyl resin on the side chains of acrylic resin can act as antifouling inhibitor even as contact killer for bio-organisms due to their rough morphologies in seawater. The relative smooth surface formed by the low elastic modulus of silicone and hydrophilic nature of the PEG segments may facilitate the release of the fouled isobornyl moieties. Thus, the combination of these synergistic effects leads to a high antifouling performance of prepared polymer coatings in seawater.

Conclusions

The IPN-FRCs could be prepared with the polyacrylic resin and hydrophilic polysiloxane resin using a so-called interpenetrating polymer network (IPN) method. The disappearance of the characteristic stretching vibration around 1650 cm^{-1} for the C=C double confirms completion of copolymerization reaction in the acrylic resins. The swelling and water contact angle measurements suggest that the IPN-FRCs are hydrophilic when submerged in water for a prolonged period and their hydrophilicity is mainly affected by the content of polysiloxane resin. The adhesion strength of

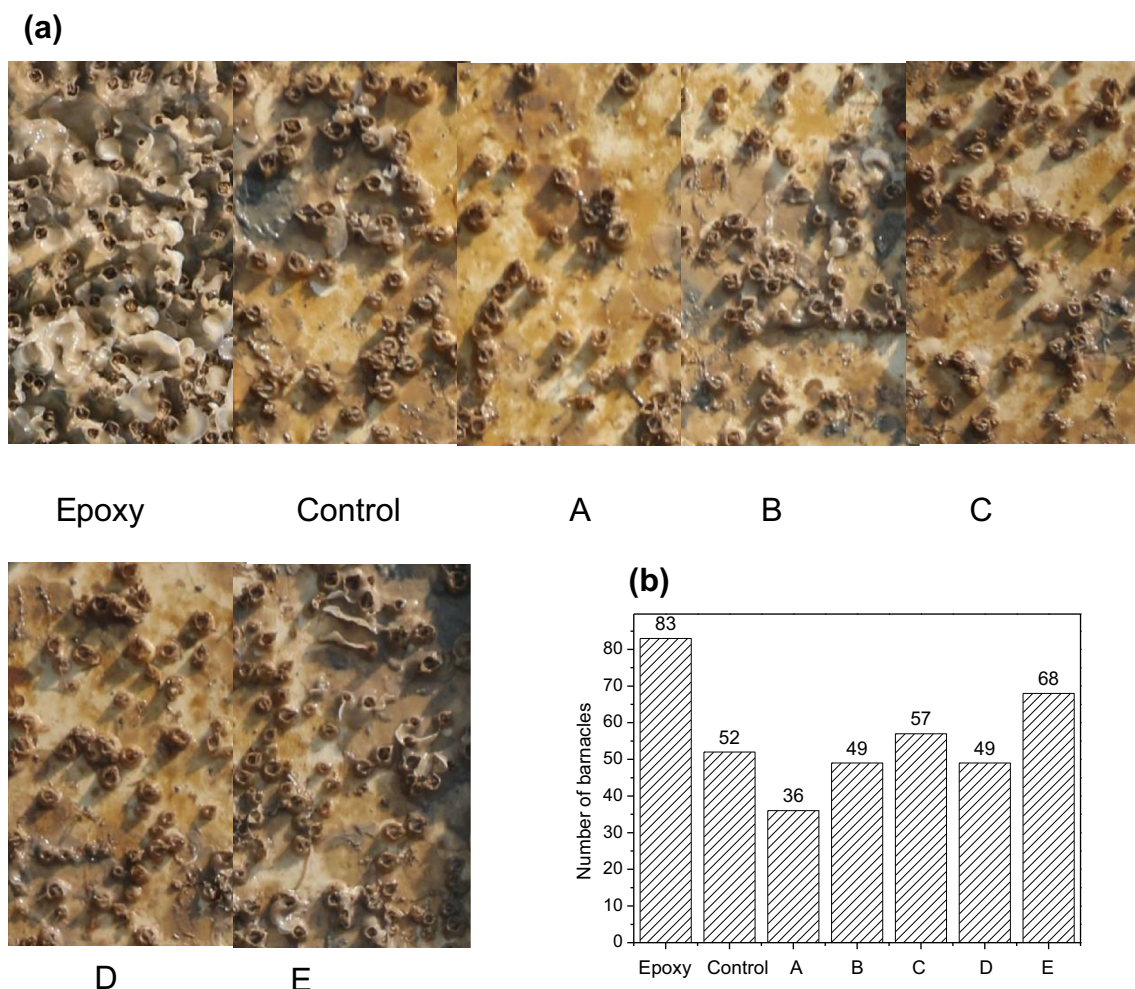


Fig. 7: Marine field test. (a) Images of surfaces of the coatings after submerged in seawater for 70 days; and (b) number of barnacles on each coating surface.

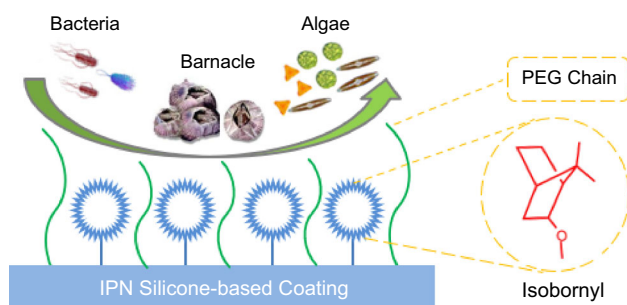


Fig. 8: Illustration of the antifouling mechanism of the applied IPN-FRCs.

these coatings on the substrate is well above the required 1 MPa, suggesting their potential long-term durability. Furthermore, the marine field test shows that these AFP coatings have improved performance against seawater bio-organism settlement, such as barnacles, when right polymer ratios were chosen. Therefore, the current work on the preparation of antifouling coatings may pave a way to introduce new

lasting, low-cost, and high-performance IPN-FRCs for marine vessels.

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Conflict of interest The author declares that there is no conflict of interest that could be perceived as prejudicing the impartiality of the research reported.

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